POKHARA UNIVERSITY

Level: Bachelor | Semester : Spring | Year : 2023
Programme: BE | Full Marks: 100
Course: Algebra and Geometry | Pass Marks: 45
Time : 3hrs.

Candidates are required to give their answers in their own words as far as practicable.

The figures in the margin indicate full marks.

Attempt all the questions.

1. a) Check consistency and solve by Gauss elimination method $x+y+z=6, x-y+z=2, 2x+y-z=1$. 7

   b) Define eigen value and vector. Find the eigen value, eigenvector and diagonalize of the matrix $A=\begin{pmatrix}2 & 1\\ 4 & -1\end{pmatrix}$ 8

2. a) Solve the linear programming problem by simplex method.[constructing duality] 7

   Minimize $Z= 2x_1+ 9x_2+ x_3$ subject to $x_1+ 4x_2+ 2x_3\geq 5$, $3x_1+x_2+2x_3\geq 4$, $x_1 \geq 0$, $x_2 \geq 0$, $x_3 \geq 0$.

   OR

   Using simplex method maximize $Z = 150x_1+300x_2$ subject to $2x_1+x_2\leq 16$, $x_1+x_2\leq 8$ *and* $x_2\leq 3.5$, $x_1\geq 0$, $x_2\geq 0$.

   b) Solve (Big M- method) Maximize $z = -3x_1 +7x_2$ subject to $2x_1 + 3x_2\leq 5$, $5x_1 +2x_2 \geq 3$, $x_2 \leq 1$ 8

3. a) State D' Alembert's Ratio test show that 7

   i. $\sum \frac{1}{n}$ is divergent.

   ii. $\sum \frac{1}{n^2}$ is convergent,

   iii. $\sum \frac{(-1)^n}{n}$ is conditional convergent.

   b) Find the center, radius and interval of convergence of the power series $\sum_{n=1}^{\infty} \frac{(x)^{2n+1}}{(-4)^n}$. 8

4. a) Define eccentricity of a conic section, and derive the equation of the ellipse in its standard form. $\frac{x^2}{a^2}+\frac{y^2}{b^2}=1$ 7

   OR

   Show that the line $lx + my + n = 0$ touches the hyperbola $\frac{x^2}{a^2}-\frac{y^2}{b^2}=1$ if $a^2l^2 - b^2m^2 = n^2$. Also find the point of contact.

   b) Show that the equations: $16x^2 - 24xy + 9y^2 - 104x - 172y + 44 = 0$ represents a parabola. Also find the equation of axis and vertex. 8

5. a) Define scalar triple product and reciprocal vectors. Find the reciprocal vector of 7

   $\vec{2i}+\vec{3j}-\vec{k}$, $\vec{i}-\vec{j}-\vec{2k}$, $\vec{-i}+\vec{2j}+\vec{2k}$.

   b) Find the distance of the point (3,-4,5) from the plane $2x+5y+6z=16$ measured along a line with direction cosines proportional to 2,1,-2 8

6. a) Find the shortest distance between the lines $\frac{x+3}{-4}=\frac{y-6}{3}=\frac{z}{2}$ and $\frac{x+2}{-4}=\frac{y}{1}=\frac{z-7}{1}$ Also Find the equation of shortest distance 7

   b) Find the equation of the sphere for which the circle $x^2+y^2+z^2+7y-2z+2=0, 2x+3y+4z=8$ is a great circle. 8

7. Attempt all the questions 2.5×4

   a) Show that $[\vec{a}\times\vec{b}\ \ \vec{b}\times\vec{c}\ \ \vec{c}\times\vec{a}]=[\vec{a}\ \vec{b}\ \vec{c}]^2$

   b) Find the centre the hyperbola: $5x^2-4y^2+20x+8y=4$.

   c) Show that the mapping $T: R^2\rightarrow R^2$ define by $T(x, y) = (x, x+y)$ is linear.

   d) Plane through the OX and OY include an angle α. Show that their lines of intersection lie on the cone is $(z^2+x^2)(y^2+z^2)\cos^2\alpha=x^2y^2$.